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Title of chapter: Chapter 10 - Fairness

Summary:

This chapter explores how fairness plays a critical role in building trustworthy machine learning (ML) systems. It begins by emphasizing that even accurate models can be unfair if they disproportionately affect specific groups, which is frequently due to biases in data collection, feature selection, or decision-making. The chapter highlights that fairness is not a single, universal idea; rather, it is determined by the designer's world view. These worldviews, like "what you see is what you get", "we're all equal" and "what you see is what you get – but we'll fix it later" influence how fairness is measured and implemented. The chapter also presents three major fairness metrics: independence/statistical parity (model's predictions should be independent of protected attributes), separation/equalized odds (model's predictions should have equal error rates across groups, both in terms of true positive and false positive rates), and sufficiency/calibration (given a predicted score, the likelihood of the outcome should be equal across all groups), each of which have a unique interpretation of what it means for an algorithm to treat groups fairly. The chapter goes on to identify where bias can enter the ML lifecycle and presents practical methods to mitigate bias at different stages of the ML pipeline: pre-processing, in-processing, and post-processing. Pre-processing methods introduced are data augmentation (uses synthetic counterfactual data to balance representation and simulate a world without bias), reweighing (assigns higher weights to underrepresented or disadvantaged groups during training to achieve statistical parity), massaging (selectively changes some labels to equalize outcomes across groups, e.g., flipping labels near the decision boundary), and fair score transformer (pre-processing version: adjusts model scores so that the distribution across groups is fair without altering the model structure). In-processing methods modify the learning algorithm itself using techniques such as prejudice remover (adds a fairness regularization term that discourages correlations between predictions and protected attributes), recent in-processing algorithms (introduce fairness constraints directly in the optimization objective, covering multiple fairness criteria such as independence, separation, and sufficiency, and causal regularizer (a more advanced technique that removes unfair causal relationships between protected attributes and predictions to achieve counterfactual fairness). Some post-processing methods introduced were group fairness post-processing (adjusts decision thresholds or flips labels within groups to equalize outcomes or error rates), individual and group fairness post-processing (selects predictions near the decision boundary for flipping to minimize both group-level and individual unfairness), and fair score transformer (post-processing version: transforms the predicted scores into fairer ones without needing the protected attribute, making it an excellent first choice when only model outputs are available).

The chapter then summarizes the decision framework for choosing mitigation strategies as: decide where in the pipeline you can intervene (pre, in, or post), then decide which worldview fits your problem ("we're all equal," etc.), and then check availability of protected attributes in deployment data. The chapter concludes by stressing that fairness must be evaluated in the construct space (a bias-free environment), not the biased prepared data space,

and that fairness should be treated as both a technical and ethical issue. It ends with a broader reflection on representational harms and the importance of considering historical and social inequities, not just mathematical metrics when designing fair ML systems.

What are two points that you learned after reading this chapter?:

- Fairness in ML may be handled from different worldviews, and the selected worldview determines whether fairness definitions and mitigation measures are appropriate.
- There are three fundamental types of intervention to reduce bias: before, during, or after model training, and each has its own benefits based on control over data and models.

What are two questions that you have about this chapter?:

- How can practitioners balance different fairness metrics (e.g., independence and sufficiency) when they cannot all be satisfied at once?
- How can fairness be quantified reliably in real-world systems when the construct space (a truly unbiased reference) is unknown or difficult to approximate?

Do you have some initial thoughts or insights to approach the questions?:

- A possible approach is to prioritize fairness metrics depending on the stakeholders' goals and the application's ethical priorities. For example, healthcare models may prioritize equal error rates (separation) over statistical parity (independence).
- I think techniques like data augmentation or simulation of counterfactual data could help approximate the construct space. Additionally, incorporating domain experts and diverse user feedback might provide a more realistic understanding of fairness in deployment environments.